THE COMPOSITION OF TWO AFFINE MORPHISMS IS AFFINE

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 1969-1972 and proof span 1974-1979 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The delivered statement is the author's one-sentence formulation; nothing was added to it, in particular no hypothesis was inserted.

Allowed standard prerequisites (editorial).

- A morphism of schemes is affine if and only if the inverse image of every affine open of the target is affine. This is the criterion applied to g at proof line 1976 and to f at proof lines 1976-1978. Supplies the single condition that has to be checked for the composite, and is applied twice in the counted proof.
- Preimages compose: the inverse image under a composite is the inverse image of the inverse image. Used at proof line 1978 to rewrite the preimage under g composed with f. Identifies the open set whose affineness has to be established for the composite.

Lemma 0.1. *The composition of affine morphisms is affine.*

Proof. Let $f : X \to Y$ and $g : Y \to Z$ be affine morphisms. Let $U \subset Z$ be affine open. Then $g^{-1}(U)$ is affine by assumption on g. Whereupon $f^{-1}(g^{-1}(U))$ is affine by assumption on f. Hence $(g \circ f)^{-1}(U)$ is affine. □

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